28/10/26 - TASK SCHEDULING.

$c_{ik,t,w} \in \{0,1\}$ $\quad c_{ik,t,w} = 1$ AT ik, RUN TASK t ON WORKSTATION w.

$W_t \subseteq W$ SET OF WORKSTATIONS THAT CAN RUN t.

CONSTR.

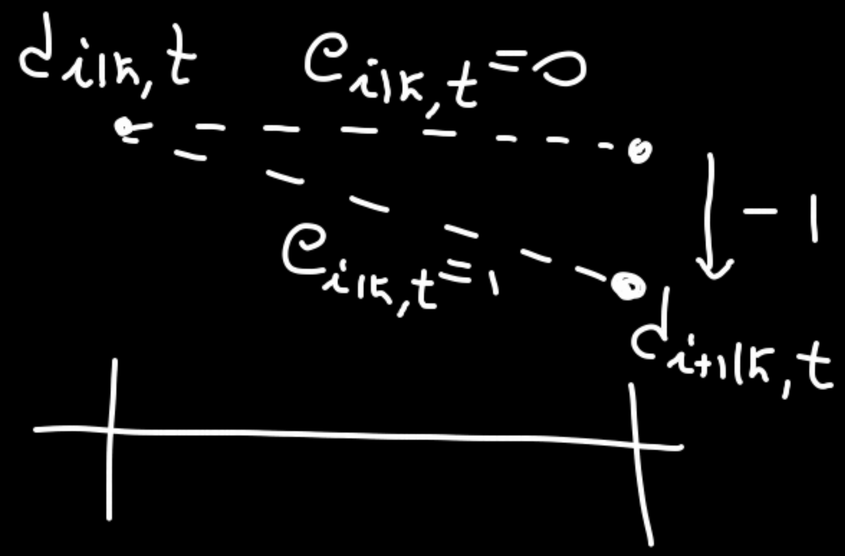
$$\underbrace{\sum_{w \in W_t} c_{ik,t,w}}_{\text{(CAN)}} \leq 1 \quad ; \forall i,t$$

$$\| \; c_{ik,t}$$

TIME LEFT TO COMPLETE

d_t TOTAL TIME [NUMB. OF TIME STEPS] TO COMPLETE TASK.

$d_{i|k,t}$ // EVALUATED AT $i|k$.



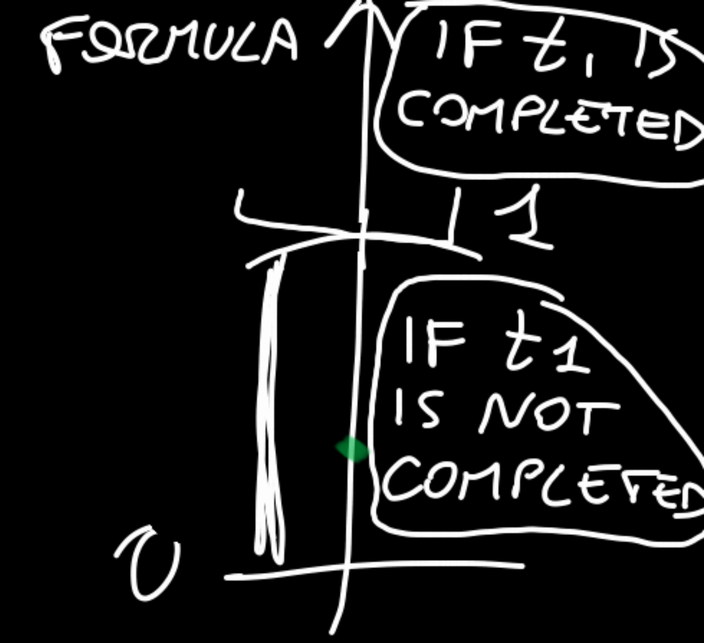
$$\begin{cases} d_{i+1|k,t} = d_{i|k,t} - c_{i|k,t}, \ \forall i,t \\ d_{0|k,t} = d_{k,t} \end{cases}$$

$d_{k,t}$ → MEASUREMENT AT k OF THE TIME LEFT TO COMPLETE TASK.

DEADLINE

$$\underbrace{d_{F_t,t} = 0}_{\text{TASK COMPLETED AT/BEFORE DEADLINE.}} \quad \forall t : F_t \leq K+N.$$

PUT A TERM ON TARGET FUNCTION IF $F_t > K+W$.



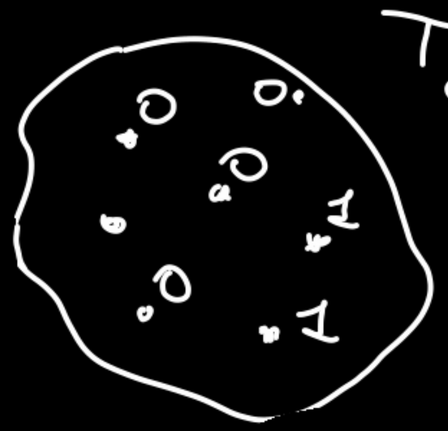


TASK t_2 CANNOT START RUN UNTIL t_1 IS COMPLETED

$$\sum_{W \in W_{t_2}} c_{ilk,t_2,W} \leq \underbrace{1 - \frac{d_{ilk,t_1}}{d_{t_1}}}_{\rightarrow \text{DEGREE OF COMPLETNESS.}} ; \forall i$$

TASKS THAT ARE MUTUALLY EXCLUSIVE





TASKS IN T_1 AND T_2 CANNOT BE RUN SIMULTANEOUSLY.

$$\sum_{w \in W_t} c_{ilk,t,w} \leq \left(1 - \frac{\sum_{t \in T_2} c_{ilk,t}}{|T_2|}\right), \quad \forall ilk, \forall t \in T_1$$

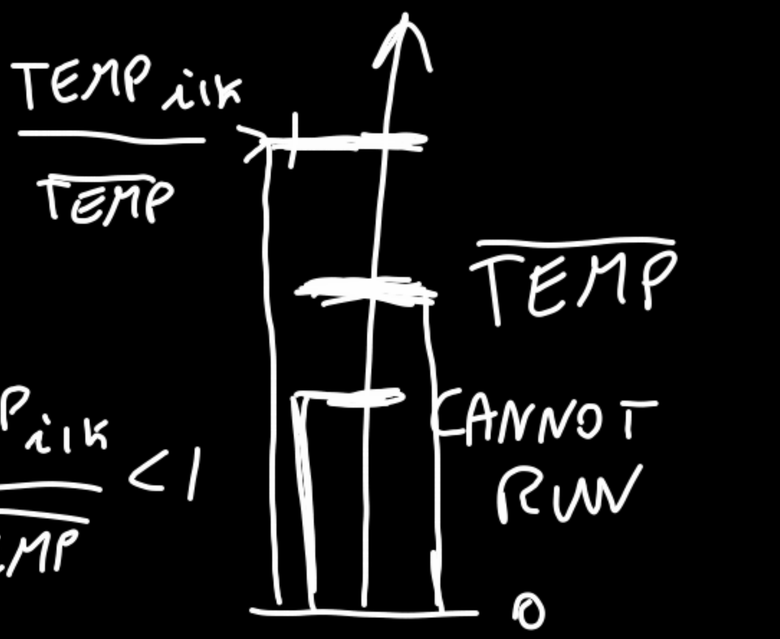
$c_{ilk,t}$: 0/1

FRACTION OF TASKS THAT ARE NOT RUNING

AT EVERY TIME ilk, EVERY TASK IN T_1 MUST NOT EXECUTE IF ONE OR MORE TASKS IN T_2 ARE EXECUTING.

CONSTRAINT ON ENVIRONMENTAL CONDITIONS

t THAT CANNOT RUN IF ABSOLUTE TEMPERATURE [KELVIN] IS BELOW $\overline{TEMP}$



WE HAVE AVAILABLE A PREDICTED TEMPERATURE, $TEMP_{ilk}$

$$0 < \frac{TEMP_{ilk}}{\overline{TEMP}} < 1$$

$$c_{ilk,t} \leq \frac{TEMP_{ilk}}{\overline{TEMP}} \quad ; \forall i$$

WORKSTATIONS

MAX/MINIMUM WORKLOAD [# OF TASKS RUNNING ON THE WORKSTATION AT GIVEN TIME].

W_w^{min}, W_w^{max} MIN AND MAX WORKLOAD FOR THE WORKST.

$$W_w^{min} \leq \sum_{t: w \in W_t} C_{ijk,t} \leq W_w^{max} \quad ; \quad \forall ijk, w$$

$\sum_{t: w \in W_t} C_{ijk,t} = W_{ijk,w}$ CURRENT WORKLOAD OF OF WORKSTATION W.

RESOURCES. (TOOLS / PEOPLE / NON CONSUMABLE RES.)

$\forall t$: $R^{\#}_{t,r}$ NUMBER OF RESOURCES (TOOLS / PEOPLE) OF TYPE r NEEDED TO RUN t.

TASK LEVEL

1	0	3	0	0	0	5

ASSEMBLY LINE LEVEL

$R^{Q}_{i,k,r}$

30	5	1000	2	20	23	100

→ NUMBER OF RESOURCE ITEMS OF TYPE r AVAILABLE AT TIME i,k.

RESOURCE CONSTRAINT.

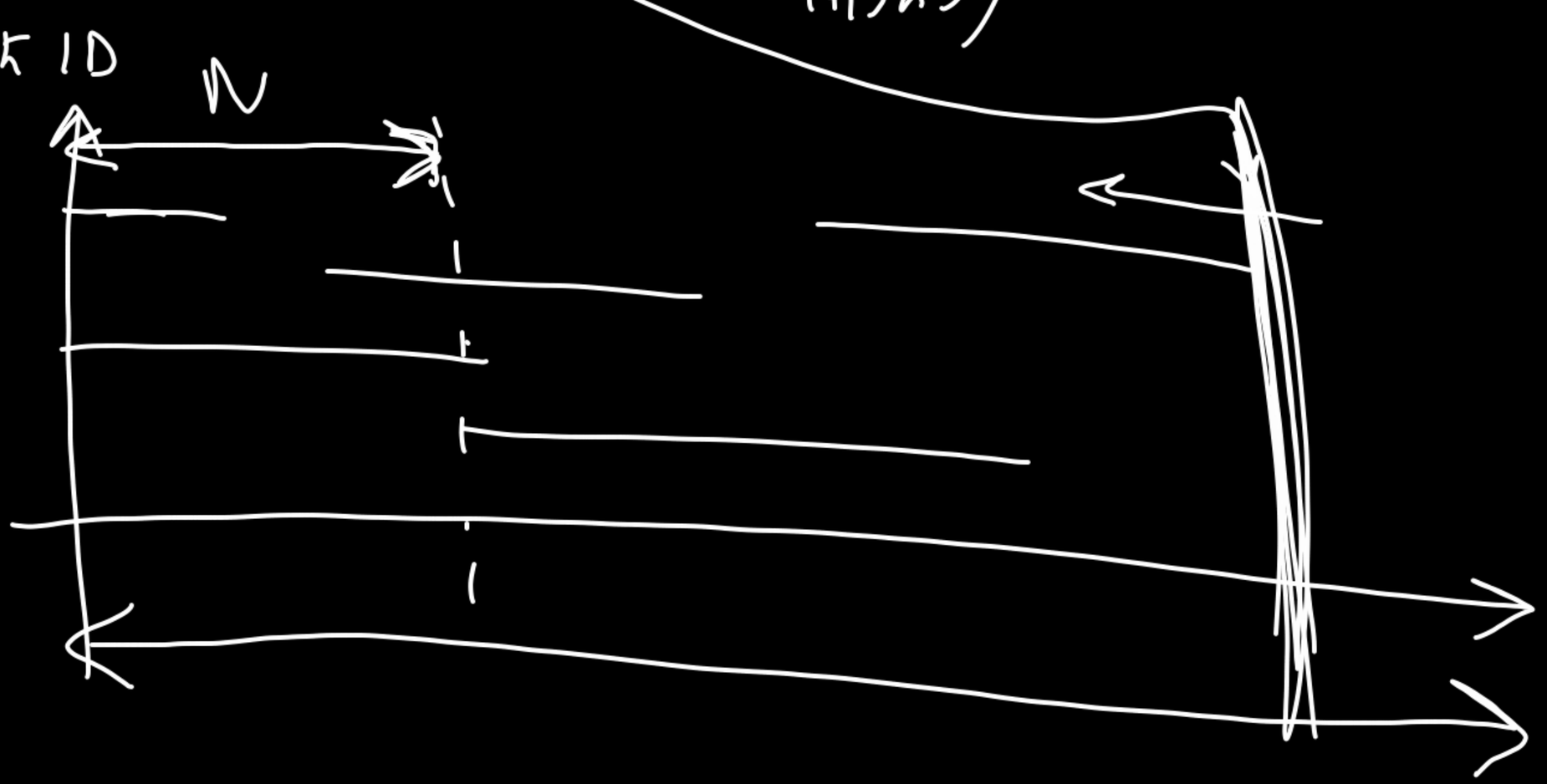
AT EVERY TIME, FOR EVERY RES. TYPE, THE TOTAL NUMBER OF ITEMS OF THAT TYPE USED BY THE RUNNING TASKS SHOULD BE LOWER OR EQUAL THAN THE AVAILABLE ONES.

$$\underbrace{\sum_{t} R^{\#}_{t,r} \cdot c^{0/1}_{i_k,t}}_{\#\text{ OF RES. OF TYPE } r \text{ USED BY ALL TASKS AT } i_k} \leq R^{a}_{i_k,r} \quad , \forall i_k, r(\text{RES. TYPE})$$

TARGET FUNCTION

MAIN GOAL:

(1) MINIMIZE MAKESPAN (TOTAL TIME TO COMPLETE TASKS)

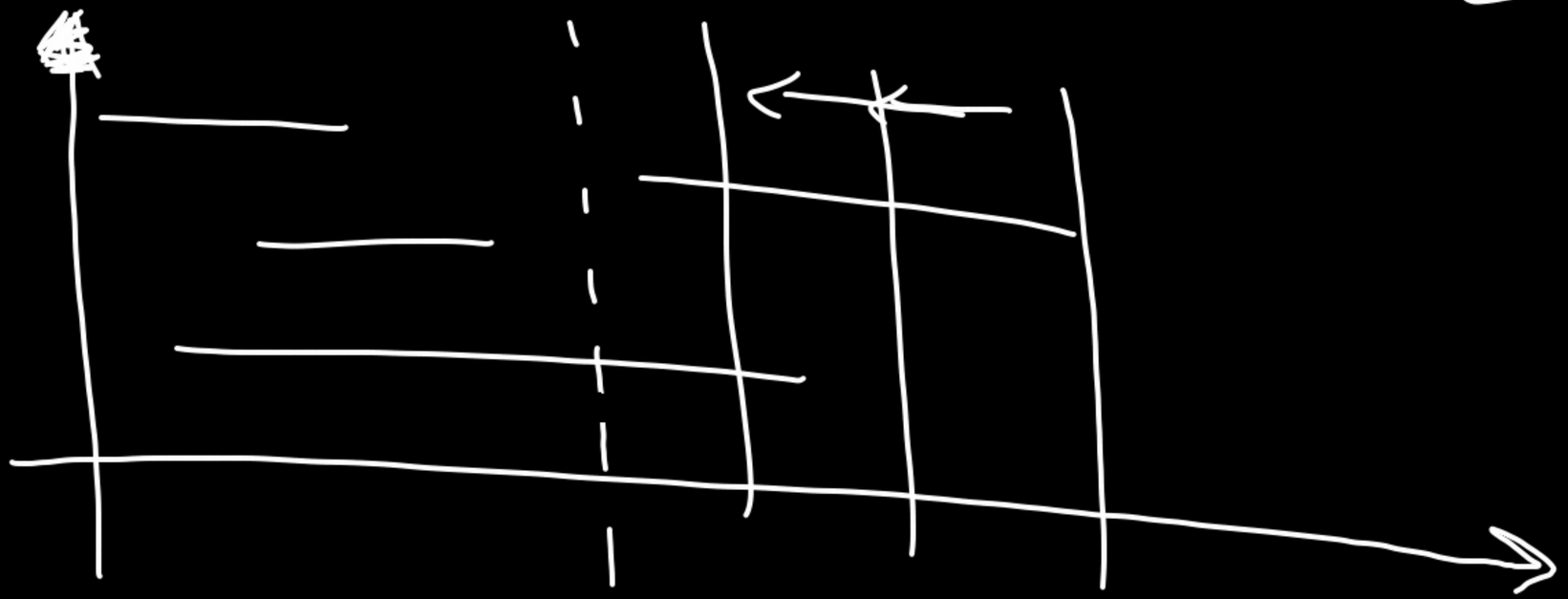


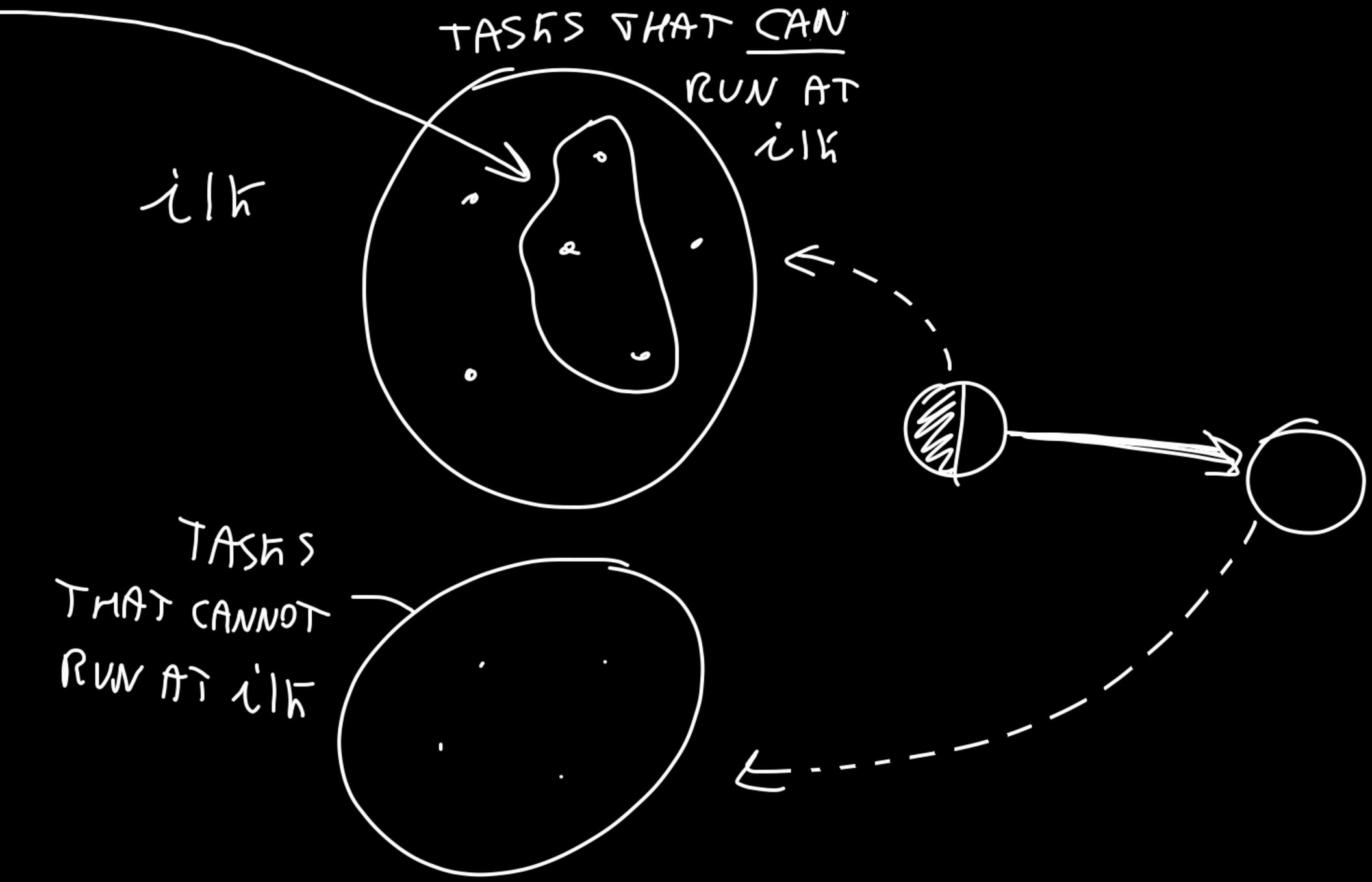
TARGET FUNCTION THAT PUSHES TASKS TO FINISH AS FAST AS POSSIBLE

$$J = \sum_{i=0}^{N-1} \sum_{t} \left(d_{ilk,t} \right)^2 + \sum_{t} W_t^f \left(d_{Nlk,t} \right)^2$$

RL

TERMINAL COST.




TASKS THAT CAN RUN AT ilk
ilk
TASKS THAT CANNOT RUN AT ilk

REFINEMENTS

$$J^{W\text{-BALANCING}} = \sum_{i=0}^{W-1} \sum_{w} \left(w_{i|k,w} \right)^2$$

↑

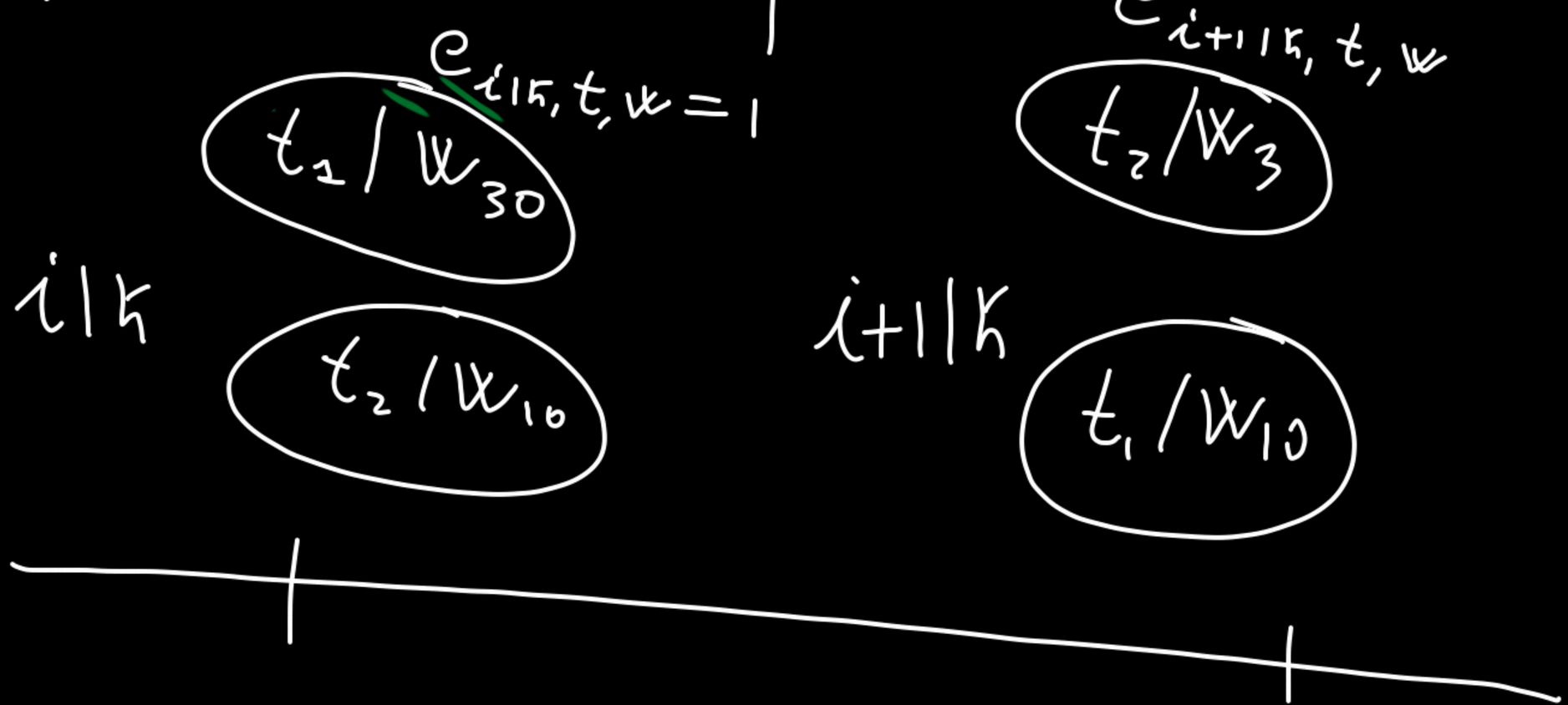
SMALL VALUE

RES. ASSIGNMENT $\quad a_{r,i|k,t} \in \{0,1\}$

MODEL TASKS INTERRUPTIONS

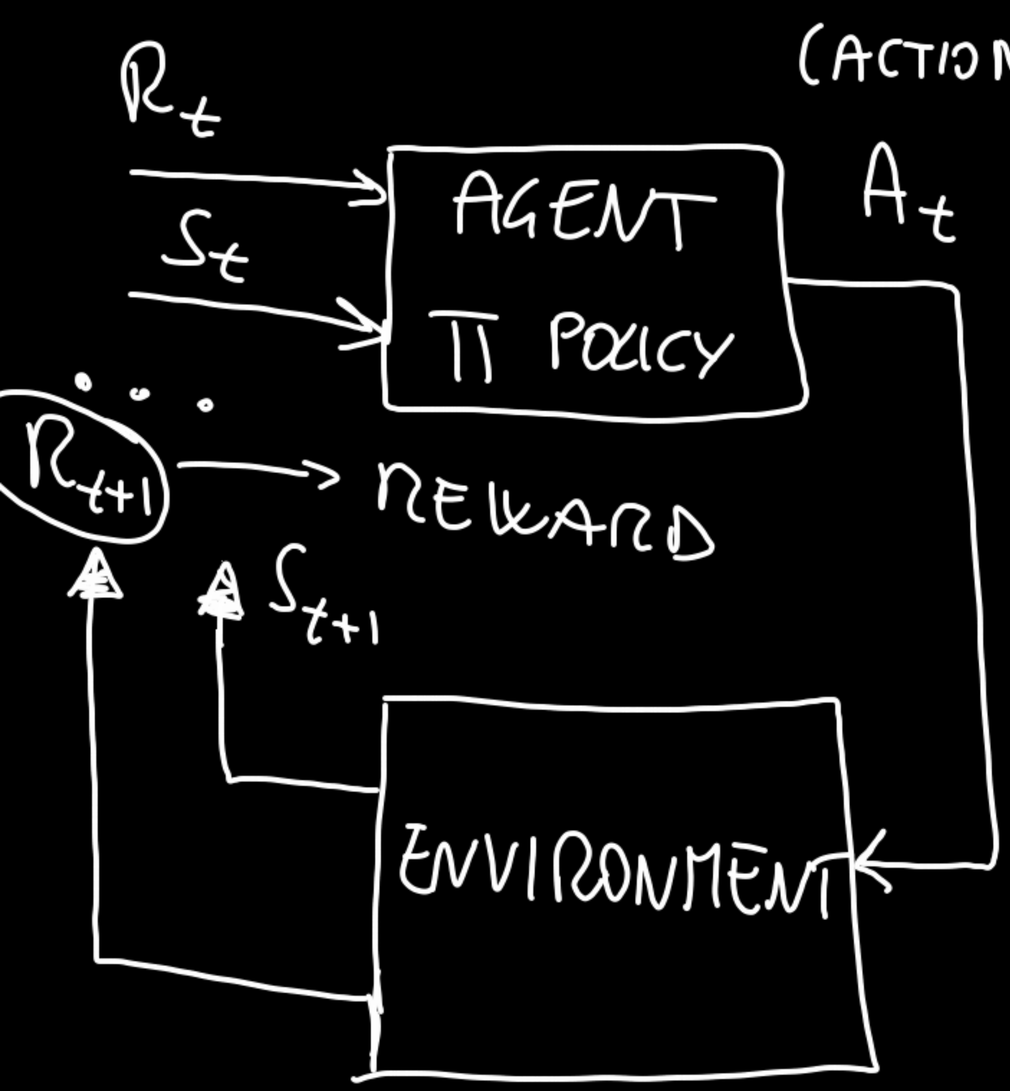
$$START_{i|k,t} \in \{0,1\} \; ; \quad c_{i|k,t} - c_{i-1|k,t} \leq START_{i|k,t}$$

AT EVERY TIME, MPC SHOULD CHANGE A SCHEDULE ONLY IF REALLY NECESSARY (PREVENT UNFEASIBILITY OR TO IMPROVE PERFORMANCE)

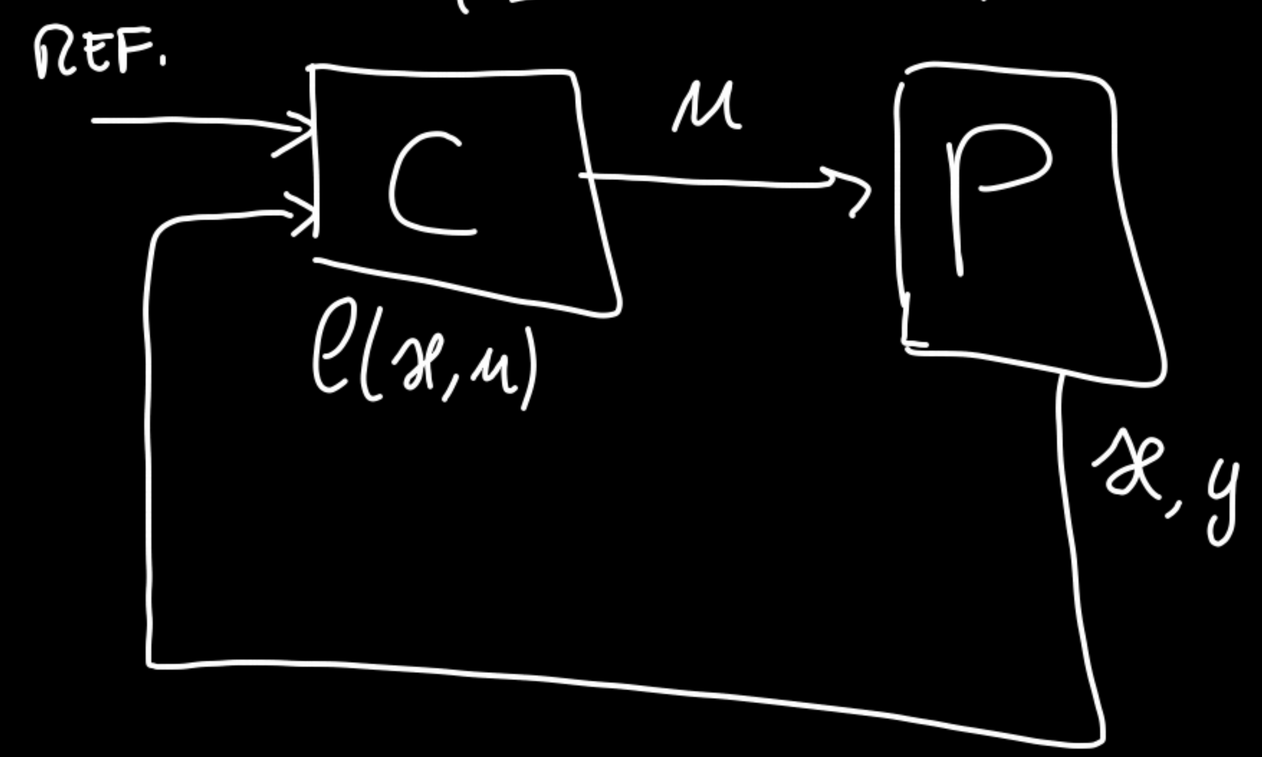


REINFORCEMENT LEARNING (RL)

RL (MAX)



MPC / CONTROL (MIN. COSTS)



REWARD $\sim l(x, u)$

REWARD VS RETURN
(Σ REWARDS)

GOAL OF RL IS TO ITERATIVELY LEARN OPTIMAL POLICIES:

POLICES THAT MAXIMIZE THE TOTAL, LONG TERM,
COLLECTED REWARD (RETURN)

$$\max \sum_{t} R_t$$

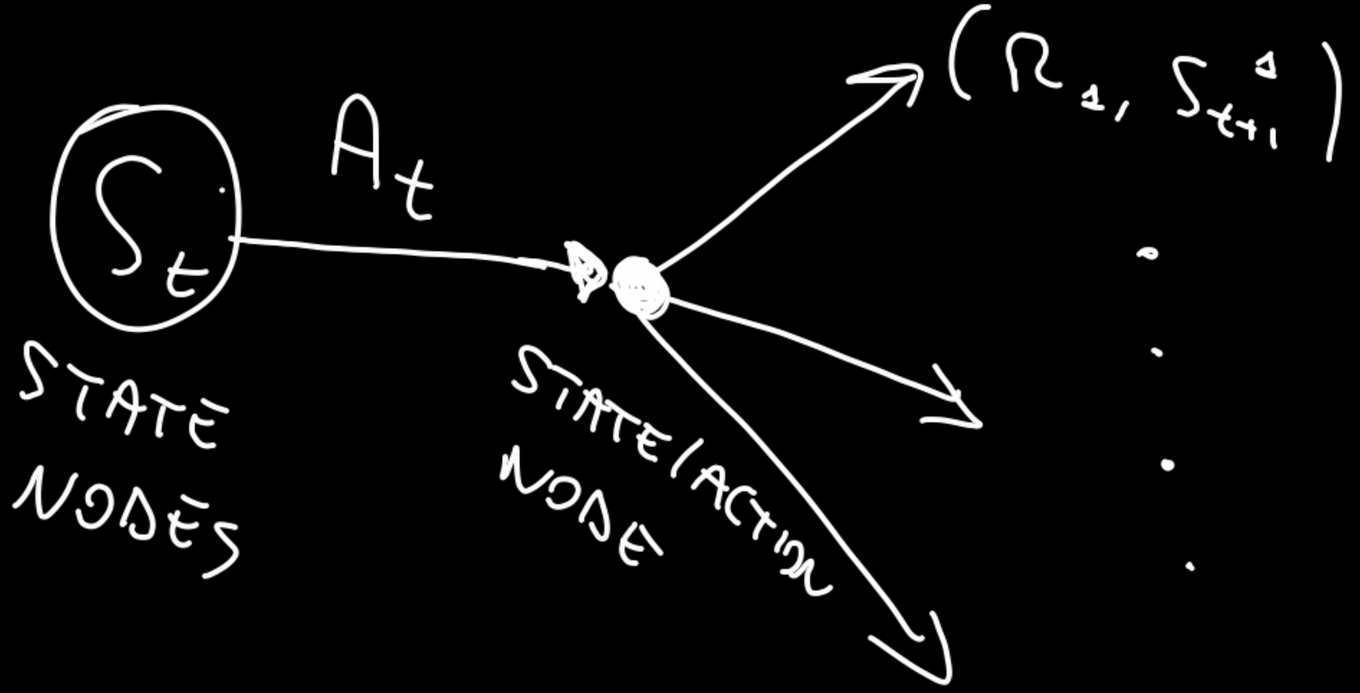
RETURN



ENVIRONMENT CAN BE PROBABILISTIC.

S_t

STATE IN RL IS EVERY USEFUL INFO
TO DETERMINE CURRENT CONFIGURATIO
OF ENVIRONMENT



POLICY FUNCTION

$\Pi(s) = a$

- <u>DETERMINISTIC</u>. $\Pi : S \longrightarrow A$

  MAPS STATES TO ACTIONS.



DISCRETE ACTIONS AND STATE SPACES.

- <u>PROBABILISTIC POLICY</u>:

$\Pi : S \times A \longrightarrow [0, 1]$

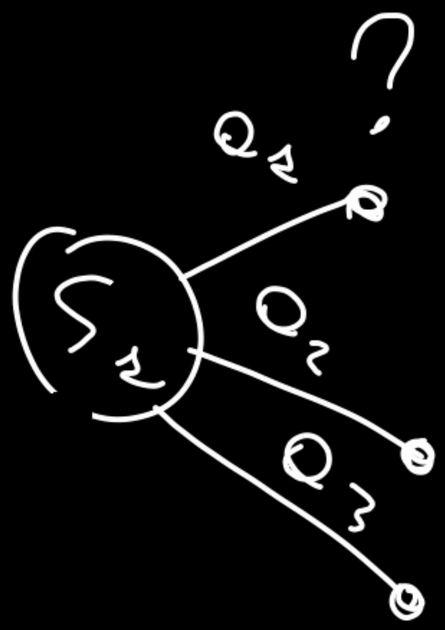
TELLS THE PROBABILITY OF TAKING A GIVEN ACTION WHEN IN A GIVEN STATE.

EX. **PROB.**

$\Pi(S_1, Q_1) = 0.2$

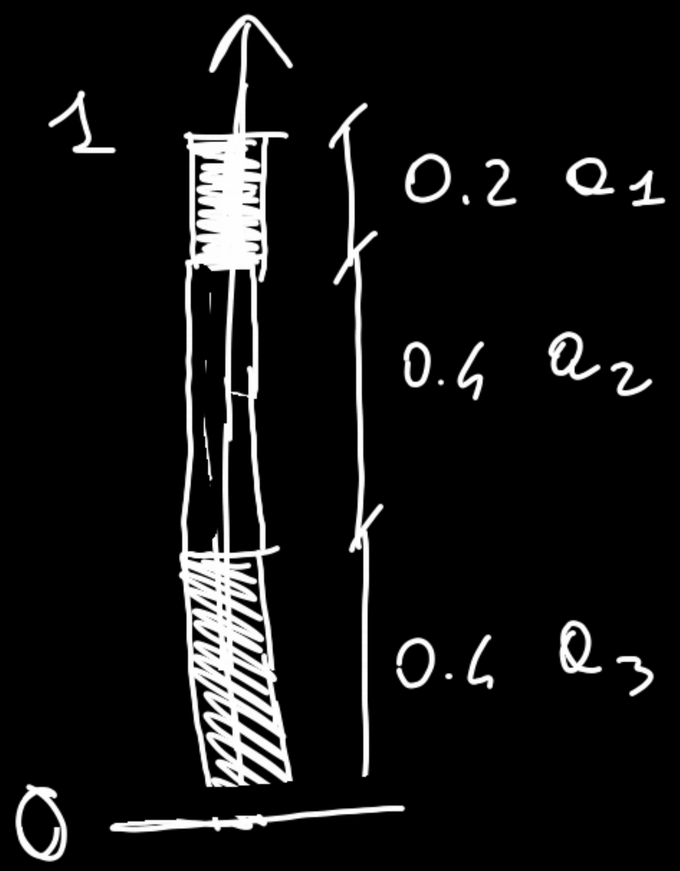
$\Pi(S_1, Q_2) = 0.4$

$\Pi(S_1, Q_3) = 0.4$



DET.

$\Pi(S_1) = Q_2$



VALUE FUNCTION(S)

STATE VALUE FUNCTION $V : S \longrightarrow \mathbb{R}$

$V_{\pi}(s)$ IS THE TOTAL EXPECTED RETURN ($\sum_t$ OF FUTURE REWARDS) STARTING FROM STATE s, AND FOLLOWING POLICY π THEREAFTER.